

Blaze Web Works Internship Programme

Remote Internship Preparation and First Week Training Plan

Dear CTU Bloemfontein Interns,

As this internship is a remote position, all sessions and communication will be handled virtually.

Because some students are part-time and others are full-time, we will not rely on one fixed meeting time for everyone. Instead, we will arrange informal talks, check-ins, and guidance sessions based on availability. The goal is to keep the internship practical, flexible, and manageable while still giving everyone meaningful exposure to real-world project work.

Please make sure you communicate clearly about your availability, especially if you are working, studying, or writing exams.

1. What You Need to Prepare

Before the first week begins, please prepare the following.

Personal Information to Provide

Each student should provide:

- Full name and surname
- Year of study
- General weekly availability
- Exam dates or study commitments, if applicable
- GitHub username, if you already have one
- Tools or technologies you have used before

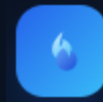
This information will help us allocate you to the correct project group and plan tasks around your availability.

2. Remote Internship Format

The internship will be handled remotely.

This means:

- Meetings will be arranged when needed



- Informal discussions will happen based on availability
- Students will receive tasks and learning material digitally
- Progress updates will be shared online
- Questions should be asked as early as possible on the FAQ group
- Work should be documented clearly
- Software work will use GitHub for version control

You are expected to take responsibility for your communication. If you are stuck, unavailable, unsure, or delayed, please communicate early.

3. Tools and Accounts to Prepare

All Students

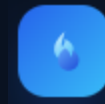
All interns should prepare:

- A working email account
- A GitHub account
- Access to the agreed communication platform
- Access to shared documentation folders when provided
- A reliable internet connection where possible
- A notebook or digital document for your internship notes
- A folder structure on your computer for internship files

Software Developers and Software Engineers

Please prepare or install:

- Visual Studio Code
- Git
- Github Desktop
- GitHub account
- Node.js (nvm)
- Npm
- Pnpm



- Chrome
- PostgreSQL, where required
- A database tool such as pgAdmin or DBeaver, if possible
- Postman or Insomnia for API testing
- Basic terminal or command-line familiarity

The Blaze Diagnostics App uses:

- TypeScript
- Next.js
- Tailwind CSS
- PostgreSQL
- Drizzle ORM
- GitHub for version control

CAD Design Students

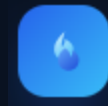
Please prepare or become familiar with:

- AutoCAD or your preferred CAD tool
- 2D drafting basics
- Floor plan concepts
- Layers, dimensions, and annotations
- Space planning
- Exporting drawings for review
- Presentation layout basics

Cyber Security Analysts

Please prepare or become familiar with:

- OWASP Top 10
- Web application security basics
- Authentication and authorization
- Role-based access control



- Password security
 - API security basics
 - Secure handling of customer data
 - GitHub secrets and why credentials should never be committed
 - Basic security reporting
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4. Learning Resources to Start With

You do not need to master everything immediately. Start with the basics and take notes on what you understand, what you do not understand, and what questions you want to ask.

Common Resources for All Students

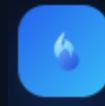
Start with:

- GitHub Docs: repositories, branches, commits, pull requests
- Basic professional communication and progress reporting
- Basic project documentation practices
- How to ask good technical or project-related questions
- How to document what you worked on each day

Software Developers and Software Engineers

Focus on:

- HTML, CSS, and JavaScript basics
- TypeScript basics
- React fundamentals
- Next.js fundamentals
- Tailwind CSS
- PostgreSQL basics
- Drizzle ORM basics
- Git and GitHub workflow
- REST API concepts
- Environment variables



- Debugging basics

Recommended resources:

- MDN Web Docs: JavaScript Guide
- TypeScript Handbook
- React official learning documentation
- Next.js official documentation
- Tailwind CSS documentation
- PostgreSQL documentation and tutorial
- Drizzle ORM documentation
- GitHub Docs: Hello World and Pull Request workflow

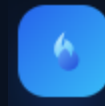
CAD Design Students

Focus on:

- AutoCAD basics
- 2D layout drafting
- Architectural floor plan concepts
- Space planning
- Movement flow between areas
- Dimensions and annotations
- Presentation drawings
- Exporting files for review

Recommended resources:

- Autodesk AutoCAD beginner tutorials
- Autodesk learning resources
- Basic architectural drafting principles
- Space planning references
- Presentation board examples



Cyber Security Analysts

Focus on:

- OWASP Top 10
- OWASP Web Security Testing Guide
- Authentication and authorization
- API security
- Role-based access control
- Secure password practices
- Secure data handling
- GitHub secret exposure risks
- Writing clear security findings

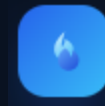
Recommended resources:

- OWASP Top 10
- OWASP Web Security Testing Guide
- GitHub security basics
- MDN HTTP basics
- PostgreSQL security basics

5. First Week Training Schedule

The first week will mainly focus on training, preparation, and understanding the technology stack. This schedule is flexible because the internship is remote and students have different availability.

You should work through the material at your own pace, but you must communicate progress and questions.



Day 1: Remote Onboarding and Availability Confirmation

Focus

Understanding the internship structure and confirming student availability.

Tasks

- Read through the welcome and orientation notes
- Confirm your contact details
- Confirm whether you are full-time or part-time
- Submit your general availability
- Submit any exam or study commitments
- Create or confirm your GitHub account
- Identify which group you belong to:
 - Software development / software engineering
 - CAD design
 - Cyber security

Outcome for the Day

By the end of Day 1, each student should have submitted their basic information, availability, and GitHub username where applicable.

Day 2: Tools and Account Setup

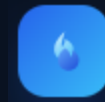
Focus

Making sure everyone has the tools needed for their project role.

Tasks for All Students

- Confirm access to the communication platform
- Confirm access to shared documents when provided
- Create a folder for internship work
- Start a personal notes document for training

Software Students



- Install Visual Studio Code
- Install Git
- Install Node.js and npm
- Confirm that Git works from the terminal
- Review GitHub basics
- Learn the difference between cloning, branching, committing, pushing, and pull requests

CAD Students

- Confirm CAD software access
- Review layers, dimensions, annotations, and layout basics
- Start researching space planning for mixed-use business centres

Cyber Security Students

- Review OWASP Top 10
- Learn basic web application security terminology
- Review why passwords, API keys, and secrets must not be uploaded to GitHub

Outcome for the Day

By the end of Day 2, each student should have confirmed which tools they can access and what still needs to be resolved.

Day 3: Web Stack Fundamentals

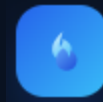
Focus

Understanding the core technology used in the Blaze Diagnostics App.

Software Students

Study:

- JavaScript basics
- TypeScript basics
- React components
- Next.js project structure
- Tailwind CSS utility classes



Key concepts to understand:

- What a component is
- What props are
- What state is
- How pages and routes work in Next.js
- How Tailwind CSS is used for styling
- How a project is run locally

Cyber Security Students

Study:

- HTTP requests and responses
- Login and session concepts
- Basic access control
- Common web application risks

CAD Students

Study:

- Synergy Inc. space requirements
- Practical flow between offices, restaurants, medical rooms, workshops, refinery areas, jewellers hub, and auction house
- Public areas versus restricted areas

Outcome for the Day

By the end of Day 3, each student should have notes on the concepts they understand and a list of questions they need help with.

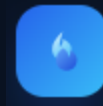
Day 4: Database, Workflow, and Security Concepts

Focus

Understanding how data and business workflows support the projects.

Software Students

Study:



- PostgreSQL basics
- Tables, rows, columns, primary keys, and relationships
- Basic SQL concepts
- What an ORM is
- What Drizzle ORM is used for
- How application code connects to a database
- Basic CRUD operations

CAD Students

Study:

- Zoning of spaces
- Operational flow
- Customer-facing areas versus staff-only areas
- Service and delivery access
- Safety and restricted areas

Cyber Security Students

Study:

- Protecting customer data
- Role-based access
- API security basics
- Database security basics
- How to write a simple security checklist

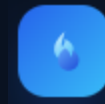
Outcome for the Day

By the end of Day 4, each student should understand how their role connects to real project workflows.

Day 5: Project Alignment and Starter Tasks

Focus

Preparing for actual project contribution.



Tasks

- Confirm project interest and preferred role
- Review which project you are most suited to
- Prepare questions for an informal discussion
- Identify what you still need help setting up
- Receive or prepare for your first small task
- Begin working on a simple starter deliverable

Possible Starter Deliverables

Software Students

- Confirm development environment setup
- Practise GitHub workflow
- Build or review a small Next.js component
- Practise Tailwind CSS styling
- Review a basic PostgreSQL table structure

CAD Students

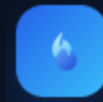
- Prepare early zoning ideas for Synergy Inc.
- Research layout requirements for mixed-use spaces
- Draft a simple conceptual layout
- Prepare notes on movement flow and space separation

Cyber Security Students

- Prepare a basic security checklist
- Identify possible risks in a workshop management system
- Identify access-control requirements
- Prepare notes on protecting customer and vehicle data

Outcome for the Day

By the end of Day 5, each student should have completed the first training phase and be ready to begin guided project work.



6. Weekly Communication Expectations

Because this is remote, communication is important.

Each student should provide regular updates covering:

- What you worked on
- What you learned
- What you completed
- What you are struggling with
- What help you need
- Your availability for the next few days

Updates do not need to be long, but they must be clear.

Example update:

“Today I worked through the GitHub basics and started learning TypeScript. I understand branches and commits better now, but I still need help understanding pull requests. I am available tomorrow afternoon after 15:00.”

7. Final Note

The first week is mainly about preparation, training, and getting comfortable with the tools and technologies.

You are not expected to know everything immediately. You are expected to learn, communicate, ask questions, and show progress.

This internship is an opportunity to gain real-world experience while still being supported as students. Use it properly, stay involved, and take responsibility for your growth.